This assignment will not be graded and is only for practice.

Key Points: 1

Scalar-valued multivariable functions: A scalar-valued multivariable function is a function $f : D \to \mathbb{R}$, where D is a domain in $\mathbb{R}^n$ where $n > 1$.
Vector-valued multivariable functions: A vector-valued multivariable function is a function $f : D \to \mathbb{R}^m$, where D is a domain in $\mathbb{R}^n$ where $n, m > 1$.

1) Which of the following functions are scalar-valued multivariable functions? ***1 point***

- [ ] $f(x, y) = xy + sin(xy)$.
- [ ] $f(x, y, z) = ln(x + y + z)$.
- [ ] $f(x, y, z) = (x^2, z, y + xz)$.
- [ ] $f(x, y) = (xy, 0)$.
- [ ] $f(x, y) = xy$.

No, the answer is incorrect.
Score: 0

Accepted Answers:

$f(x, y) = xy + sin(xy)$.
$f(x, y, z) = ln(x + y + z)$.
$f(x, y) = xy$.

2) Which of the following functions are vector-valued multivariable functions? ***1 point***

- [ ] $f(x, y) = xy + sin(xy)$.
- [ ] $f(x, y, z) = ln(x + y + z)$.
- [ ] $f(x, y, z) = (x^2, z, y + xz)$.
- [ ] $f(x, y) = (xy, 0)$.
- [ ] $f(x, y) = xy$.

No, the answer is incorrect.
Score: 0

Accepted Answers:

$f(x, y, z) = (x^2, z, y + xz)$.
$f(x, y) = (xy, 0)$.

Key Points: 2

Composition of functions: Let $D \subseteq \mathbb{R}^n$ and $f : D \to \mathbb{R}^m$ be a multivariable function. Let $g : E \to \mathbb{R}^p$ be a function on E where $Range(f) \subseteq E \subseteq \mathbb{R}^m$.

We define a multivariable function $g \circ f : D \to \mathbb{R}^p$ called the composition of f and g defined as $g \circ f(\underset{\sim}{x}) = g(f(\underset{\sim}{x}))$, for all $\underset{\sim}{x} \in D$.

3) If two functions $f : \mathbb{R}^2 \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$ are defined as $f(x, y) = (x - y)$ and $g(x) = \sin x$, then which of the following functions represents $g \circ f$? **1 point**

- $\sin x - \sin y$.
- $\sin(x - y)$.
- $x \sin x - y \sin y$.
- $\sin xy$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\sin(x - y)$.

4) If two functions $f : \mathbb{R}^2 \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}^2$ are defined as $f(x, y) = xy$ and $g(t) = (e^t, e^{-t})$, then which of the following functions represents $g \circ f$? **1 point**

- $g \circ f(x, y) = e^{xy}$.
- $g \circ f(x, y) = 1$.
- $g \circ f(x, y) = (e^{xy}, e^{-xy})$.
- $g \circ f(x, y) = (e^x, e^{-x})$.
- $g \circ f(x, y) = (1, 1)$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$g \circ f(x, y) = (e^{xy}, e^{-xy})$.

Key Points: 3

Partial derivatives: Let $f(x_1, x_2, \ldots, x_n)$ be a function defined on a domain D in $\mathbb{R}^n$. The partial derivative of f with respect to x_i is the function denoted by $f_{x_i}(\underset{\sim}{x})$ or $\frac{\partial f}{\partial x_i}(\underset{\sim}{x})$ and defined as

$$\frac{\partial f}{\partial x_i}(\underset{\sim}{x}) = \lim_{h \to 0} \frac{f(\underset{\sim}{x}+he_i)-f(\underset{\sim}{x})}{h}$$

where e_i is the unit vector $(0, 0, \ldots, 0, 1, 0, \ldots, 0)$ (at i-th coordinate there is 1, and 0 at all the other coordinates).

$$\frac{\partial f}{\partial x_i}(x_1, x_2, \ldots, x_n) = \lim_{h \to 0} \frac{f((x_1,x_2,\ldots,x_n)+h(0,0,\ldots,0,1,0,\ldots,0))-f(x_1,x_2,\ldots,x_n)}{h}$$

$$\frac{\partial f}{\partial x_i}(x_1, x_2, \ldots, x_n) = \lim_{h \to 0} \frac{f(x_1,x_2,\ldots,x_{i-1},x_i+h,x_{i+1},\ldots,x_n)-f(x_1,x_2,\ldots,x_n)}{h}$$

To calculate the partial derivative with respect to x_i, think of f only as a function of x_i while treating all other variables as constants. Then calculate it as the derivative of a function of one variable.

5) Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ such that $f(x, y) = x^2 + y^2$. Which of the following function will represent f_x or $\frac{\partial f}{\partial x}$?

1 point

- $2x + 2y$.
- $2x$.
- $2x + y^2$.
- $x + y^2$.

No, the answer is incorrect.
Score: 0

Feedback:

Treat y as constant.

Accepted Answers:

$2x$.

6) Consider the function $f : D \to \mathbb{R}$ where $D \subset \mathbb{R}^2$, defined as:

1 point

$$f(x, y) = \begin{cases} \frac{x^2 y}{x^3 + y^3} & \text{if } (x, y) = (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

What will be the partial derivative of f with respect to x at the point $(0, 0)$?

- 1.
- 0.
- 2
- Cannot be determined.

No, the answer is incorrect.
Score: 0

Feedback:

Observe that, the function is defined separately at the origin and the description is different around the origin. So in this case we have to use the definition of the partial derivative to calculate $\frac{\partial f}{\partial x}(0,0)$.

$$\frac{\partial f}{\partial x}(0,0) = \lim_{h\to 0} \frac{f(0+h,0)-f(0,0)}{h} = \lim_{h\to 0} \frac{f(h,0)-f(0,0)}{h} = \lim_{h\to 0} \frac{0-0}{h} = 0$$

Accepted Answers:

0.

Key Points: 4

Directional derivative: The rate of change of $f(x_1, x_2, \ldots, x_n)$ in the direction of the unit vector $u = (u_1, u_2, \ldots, u_n)$ is called the directional derivative and is denoted by $D_u f(x_1, x_2, \ldots, x_n)$. The definition of the directional derivative is:

$$D_u f(x_1, x_2, \ldots, x_n) = \lim_{h\to 0} \frac{f(x_1+u_1h, x_2+u_2h, \ldots, x_n+u_nh) - f(x_1, x_2, \ldots, x_n)}{h}$$

To calculate the directional derivative of a function f in a particular direction, we have to first find the unit vector u in that direction and then compute $D_u f$.
The partial derivatives are special cases of the directional derivative where $u = e_i$ where $e_i$is i-th vector of the standard ordered basis of $\mathbb{R}^n$.

7) Consider the function $f : \mathrm{R}^2 \to \mathrm{R}$, such that $f(x, y) = x + y$. Which of the following options will denote the directional derivative of f in the direction of the vector $v = (1, 1)$? ***1 point***

[Hint: The given vector v is not a unit vector. Convert v to a unit vector $u = \frac{v}{||v||}$.]

- 2
- $x + y$
- $\sqrt{2}$
- $\sqrt{2}(x + y)$

No, the answer is incorrect.

Score: 0

Feedback:

Calculate the following:

$$D_u f(x, y) = \frac{f(x + \frac{h}{\sqrt{2}}, y + \frac{h}{\sqrt{2}}) - f(x, y)}{h}$$

Accepted Answers:

$\sqrt{2}$

Key Points: 5

Limit of a scalar valued multivariable function: If there exists some real number L such that $f(x_1, x_2, \ldots, x_n)$ approaches L as $(x_1, x_2, \ldots, x_n)$ approaches $(a_1, a_2, \ldots, a_n)$, then we write

$$\lim_{(x_1, x_2, \ldots, x_n) \to (a_1, a_2, \ldots, a_n)} f(x_1, x_2, \ldots, x_n) = L$$

For functions of one variable $f(x)$, there are only two ways in which x can approach a point a, from left or from right. But for multivariable functions, $(a_1, a_2, \ldots, a_n)$ can be approached along any curve.

If C is a curve passing through $(a_1, a_2, \ldots, a_n)$, the limit of f at $(a_1, a_2, \ldots, a_n)$ along the curve C exists and equals L if $f(x_1, x_2, \ldots, x_n)$ approaches L as $(x_1, x_2, \ldots, x_n)$ approaches $(a_1, a_2, \ldots, a_n)$ where $(x_1, x_2, \ldots, x_n)$ belongs to C.

The limit of the function at $(a_1, a_2, \ldots, a_n)$ along the curve C equals L for each curve C passing through $(a_1, a_2, \ldots, a_n)$ precisely when the limit of f at $(a_1, a_2, \ldots, a_n)$ equals L.

To show that the limit of f at $(a_1, a_2, \ldots, a_n)$ does not exist, one should either produce a curve C so that the limit of f at $(a_1, a_2, \ldots, a_n)$ along C does not exist or produce two curves so that the limits of f at $(a_1, a_2, \ldots, a_n)$ along both curves are not equal.

Continuity of a function: If limit of a function $f : D \to \mathrm{R}$, where $D \subseteq \mathrm{R}^n$, exists at a point $(a_1, a_2, \ldots, a_n)$, and the functional value and the limiting value are the same, i.e.,

$$\lim_{(x_1, x_2, \ldots, x_n) \to (a_1, a_2, \ldots, a_n)} f(x_1, x_2, \ldots, x_n) = f(a_1, a_2, \ldots, a_n)$$

then we say f is continuous at the point $(a_1, a_2, \ldots, a_n)$.

Consider the function $f : D \to \mathrm{R}$ where $D \subset \mathrm{R}^2$, defined as:

$$f(x, y) = \begin{cases} \frac{x^2 y}{x^3 + y^3} & \text{if } (x, y) = (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

Answer the following question 8 to 13 using the above information.

8) If f approaches L as (x, y) approaches to the origin along the X-axis, then find the value of L.

No, the answer is incorrect.

Score: 0

Feedback:

Along the X-axis, y = 0.

Accepted Answers:

(Type: Numeric) 0

1 point

9) If f approaches L as (x, y) approaches to the origin along the Y-axis, then find the value of L.

No, the answer is incorrect.

Score: 0

Feedback:

Along the Y -axis, x = 0.

Accepted Answers:

(Type: Numeric) 0

1 point

10) If f approaches L as (x, y) approaches to the origin along the straight line $y = x$, then find the value of L.

No, the answer is incorrect.

Score: 0

Feedback:

Along the straight line $y = x$ and $x, y \neq 0$, we have $f(x, y) = \frac{x^3}{x^3+x^3} = \frac{1}{2}$.

Accepted Answers:

(Type: Numeric) 0.5

1 point

11) If f approaches L as (x, y) approaches to the origin along the straight line $y = 2x$, then find the value of L. **1 point**

- 0.
- $\frac{1}{2}$.
- $\frac{2}{9}$.
- 1.

No, the answer is incorrect.

Score: 0

Feedback:

Along the straight line $y = x$ and $x, y \neq 0$, we have $f(x, y) = \frac{2x^3}{x^3+8x^3} = \frac{2}{9}$.

Accepted Answers:

$\frac{2}{9}$.

12) If f approaches to L as (x, y) approaches to the origin along the straight line $y = mx$, then find the value of L. **1 point**

- 0.
- $\frac{1}{m^2}$.
- $\frac{m}{1+m^3}$.
- 1.

No, the answer is incorrect.

Score: 0

Feedback:

Along the straight line $y = x$ and $x, y \neq 0$, we have $f(x, y) = \frac{mx^3}{x^3+m^3x^3} = \frac{m}{1+m^3}$.

Accepted Answers:

$\frac{m}{1+m^3}$.

13) Choose the set of correct options. ***1 point***

- [ ] $\lim_{(x,y)\to(0,0)} f(x,y)$ does not exist.
- [ ] $\lim_{(x,y)\to(0,0)} f(x,y) = 0$.
- [ ] f is not continuous at the origin.
- [ ] $\lim_{(x,y)\to(0,0)} f(x,y) = \frac{1}{2}$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\lim_{(x,y)\to(0,0)} f(x,y)$ does not exist.

f is not continuous at the origin.

14) Consider the function $f : D \to \mathbb{R}$ where $D \subset \mathbb{R}^2$, defined as: **_1 point_**

$$f(x,y) = \begin{cases} \frac{y}{x^2+y} & \text{if } (x,y) = (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$$

Choose the set of correct options.
[Hint: Approach the origin using the curves $y = x^2$ and $y = 2x^2$.]

- [ ] $\lim\limits_{(x,y)\to(0,0)} f(x,y)$ does not exist.
- [ ] $\lim\limits_{(x,y)\to(0,0)} f(x,y) = 0.$
- [ ] f is not continuous at the origin.
- [ ] $\lim\limits_{(x,y)\to(0,0)} f(x,y) = 1.$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$\lim\limits_{(x,y)\to(0,0)} f(x,y)$ does not exist.
f is not continuous at the origin.

Key Points: 6

Gradient of a function: If f is a scalar valued function defined on a domain D in $\mathbb{R}^n$, then the gradient ∇f is defined as the vector valued function $(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_n})$.
If ∇f exists and is continuous in an open ball around the point $(a_1, a_2, \dots, a_n)$, then the directional derivative of the function f at $(a_1, a_2, \dots, a_n)$ in the direction of a vector v is given by the dot product of the gradient of the function at $(a_1, a_2, \dots, a_n)$ with the unit vector in the direction of v i.e.,

$$D_v f(a_1, a_2, \dots, a_n) = \nabla f(a_1, a_2, \dots, a_n).\frac{v}{||v||}$$

15) Find the directional derivative of the function $f : \mathrm{R}^2 \to \mathrm{R}$ such that $f(x, y) = 3x^2 - 2xy$ in the direction of the vector $(3, 4)$.

1 point

- $2x - \frac{6y}{5}$.
- $10x - 6y$.
- $10x$.
- $2x$.
- $-\frac{6y}{5}$.

No, the answer is incorrect.
Score: 0

Accepted Answers:

$2x - \frac{6y}{5}$.

16) Find the directional derivative of the function $f : \mathrm{R}^2 \to \mathrm{R}$ such that $f(x, y) = 3x^2 + 2y^2$ at the point $(1, 1)$ along the direction of it's gradient vector $\nabla f(1, 1)$.
[Hint: First find out the unit vector in the direction of the vector $\nabla f(1, 1)$.]

1 point

- 2
- $\sqrt{13}$
- $2\sqrt{13}$
- $2\sqrt{2}$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$2\sqrt{13}$

Your score is: 0/16